



CYCLISTIC BIKE-SHARE ANALYSIS

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BUSINESS TASK

Analyze how **Casual Riders** and **Annual Members** use Cyclistic bikes differently to support marketing strategies that increase annual memberships.

DATA SOURCES AND META DATA

- The Analysis uses **Cyclistic historical Dataset** stored in CSV. The dataset contains ride-level records including timestamps, rider categories station information, and ride duration.
- Initial Data Exploration revealed **several quality issues** such as Zero time duration rides, and invalid timestamps that require cleaning before Analysis.

DATA CLEANING PROCESS

01



Convert Timestamps

Converted started_at and ended_at to proper datetime format to ensure consistency.

02



Calculate Ride Duration

Calculated ride_length as the difference between end time and start time.

03



Remove Invalid Rides

Removed rides with negative or zero duration and records where start time was greater than end time.

04



Filter Suspicious Records

Filtered records with missing values, incorrect data or suspicious station information.

05



Check Duplicates

Checked for duplicate trip_id records and removed duplicate entries.

06



Standardize Rider Category

Standardized rider type values to “Member” and “Casual” for uniformity.

07



Create Time-based Variables

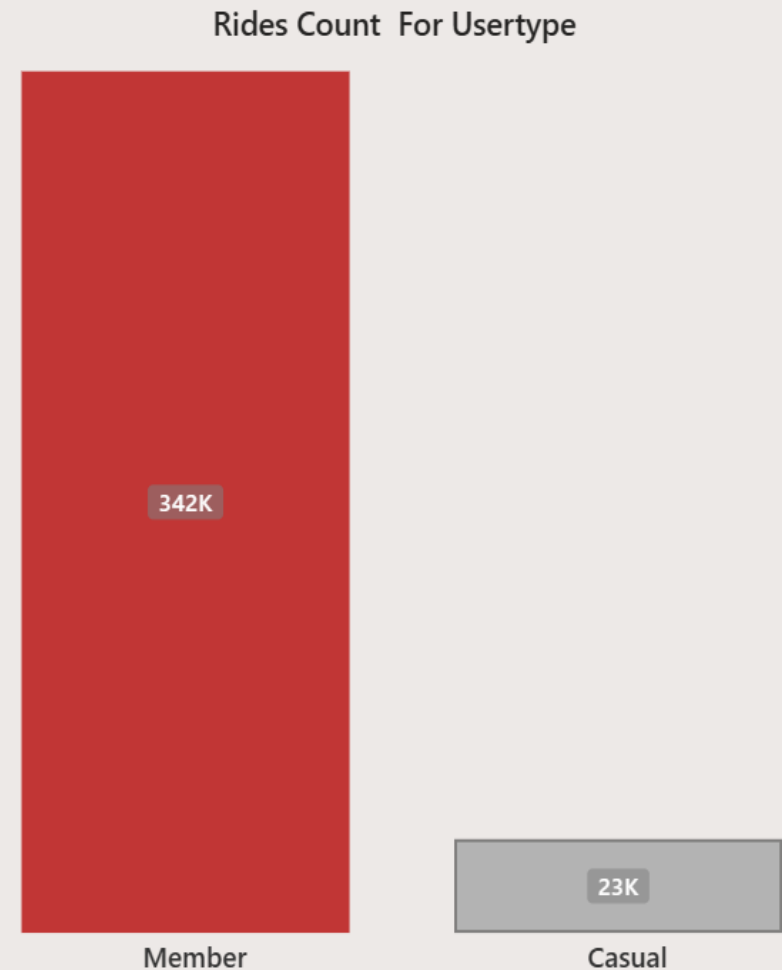
Extracted additional variables such as day_of_week, day_name, month, and hour for analysis.

SUMMARY OF ANALYSIS

- The Analysis found clear behavioral differences between customer and subscriber groups.
- Members completed significantly **more rides**, whereas casual riders had much longer average ride durations, **indicating recreational usage**.

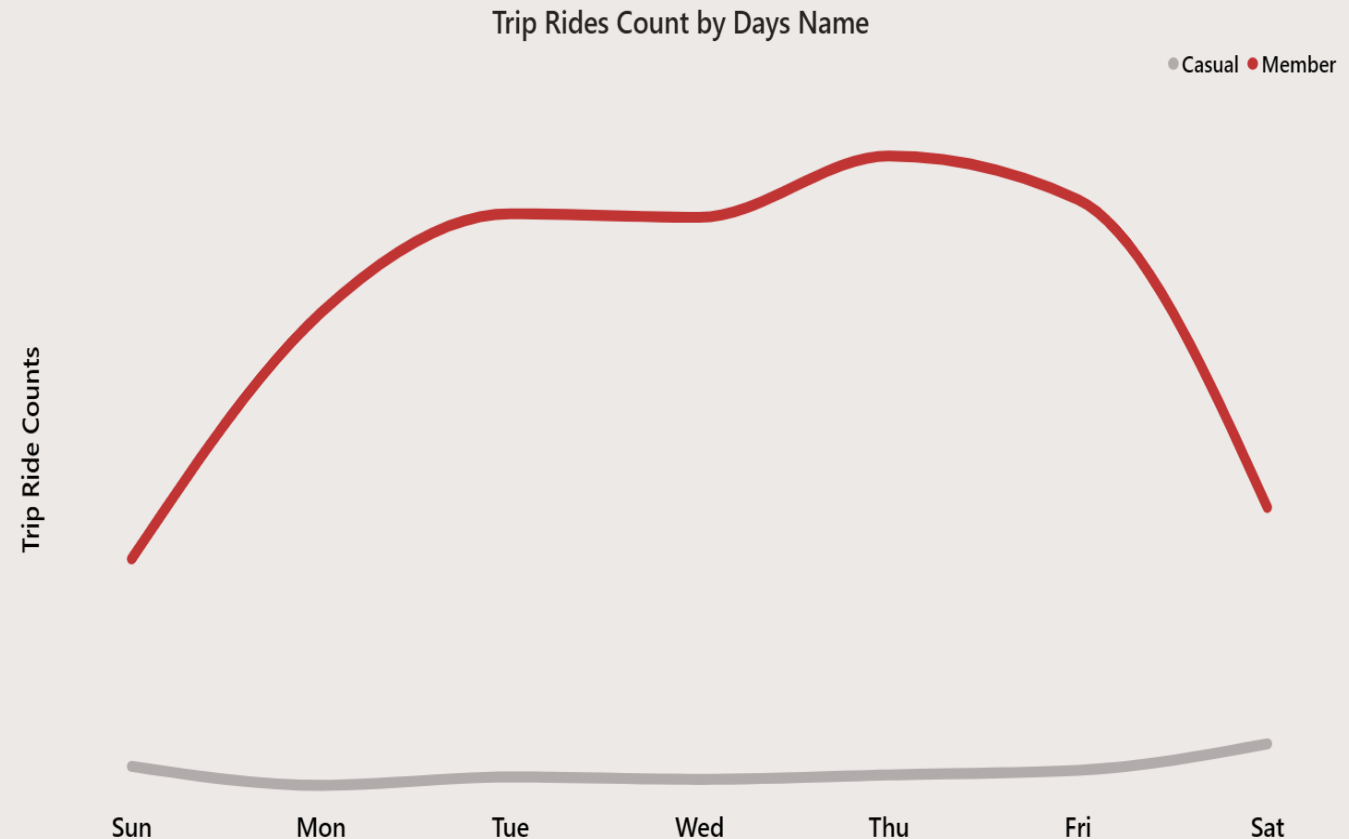
RIDE COUNT ANALYSIS BY USER TYPE

- ❑ The chart shows that members completed approximately **342K rides**, while casual riders completed only **23K rides**.
- ❑ This indicates that members use Cyclistic bikes far more frequently than casual riders, suggesting **regular commuting behavior**.



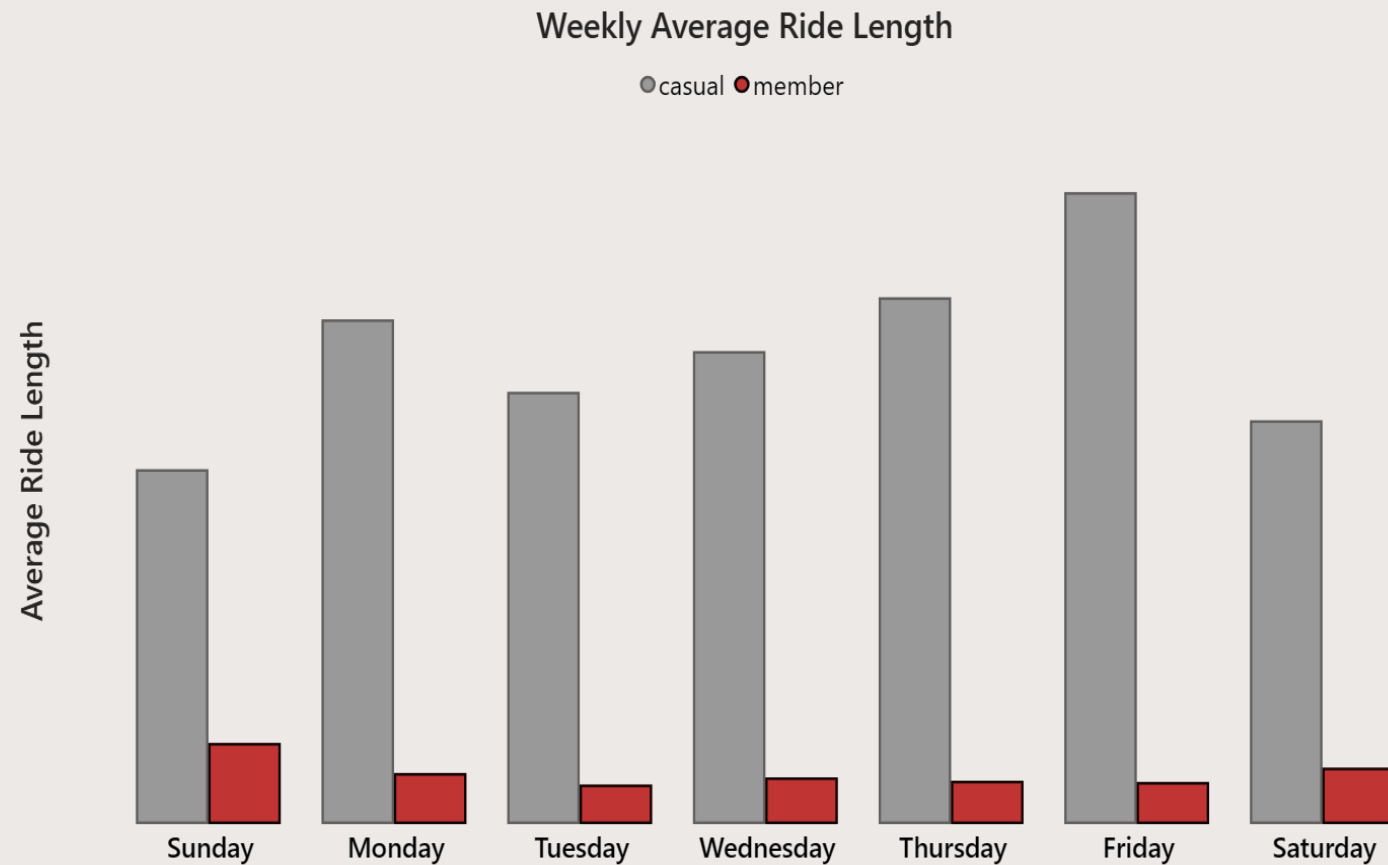
WEEKLY RIDE COUNT COMPARISON

- ❖ The chart shows that members consistently complete more rides throughout the week, **peaking on Thursday** and **least on weekends**.
- ❖ Casual riders record fewer trips overall, with slightly higher activity during weekends, indicating **recreational usage**.



WEEKLY RIDE COUNT COMPARISON

- ❖ Casual riders spend significantly **more time per ride** across the week, whereas members maintain shorter and **consistent ride durations**, reflecting recreational versus daily commuting behavior.



RECOMMENDATIONS



1. Weekend Membership Campaigns

- Cyclistic should launch special weekend membership promotions and discounted annual plans to encourage casual riders to become members.



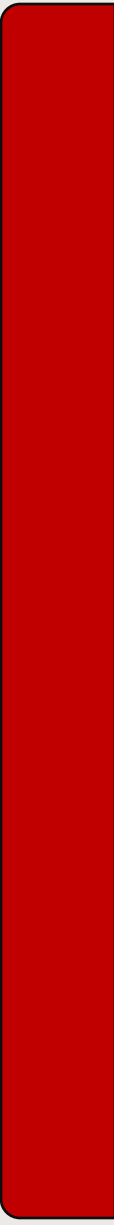
2. Promote Cost Savings for Frequent Casual Riders

- Cyclistic should identify frequent casual riders and send personalized messages showing how much money they could save by purchasing an annual membership.



3. Digital Marketing Focused on Leisure Riders

- Cyclistic should create targeted social media and mobile app campaigns highlighting the convenience, flexibility, and savings of annual memberships for leisure riders.



THANK YOU!

Thank you for your attention